Lightweight Object Detection Network With Spatial-Channel Attention and Stair Fusion Structure for Remote Sensing Open-Pit

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***Abstract*— Effective monitoring methods for open-pit mining areas are important safeguards for mineral resource and environmental protection. Due to the complex surface environment of mining areas and the difficulty in defining the edges, conventional methods are difficult to detect accurately. Furthermore, previous deep-learning-based approaches have high memory and computational costs. Therefore, we propose a novel lightweight network with dual-attention and stair fusion structure called DASFNet for pixel-level object detection in remote sensing mining areas. First, we design a neighbor feature aggregation module (NFAM) to integrate adjacent features, which can enhance the representation of features at each level. Second, an edge feature compensation module (EFCM) is developed to strengthen edge information and improve the detection performance of mining area edges. In addition, a cross-attention fusion module (CAFM) is developed to weight features from adjacent levels, with the aim of better aggregating features from high to low levels. Finally, we create and introduce a dataset called CIGMR for open-pit mining area detection. To our knowledge, CIGMR is the first dataset for open-pit mining area detection. At the same time, by applying fog simulation algorithms to the CIGMR dataset, we obtained the CIGMR(fog) dataset. Extensive experiments demonstrate that DASFNet is superior to six state-of-the-art (SOTA) methods on the CIGMR and CIGMR(fog) datasets, which has fewer parameters and lower computational cost.**

***Index Terms*— Deep learning, dual-attention, lightweight object detection, remote sensing open-pit, stair fusion.**

I. Introduction

MINERAL resources are an important part of natural resources [1], and effective mineral resources management methods are essential prerequisites for ensuring orderly and sustainable exploitation, maintaining a healthy ecological environment and steadily promoting the construction of ecological civilization [2]. As important carriers of mineral resources, open-pit mines provide essential raw materials for urban construction. However, driven by profit, illegal mining and unauthorized extraction occur frequently, severely harming national interests and causing irreparable damage to the ecological environment [3]. Therefore, establishing a rapid and effective information collection platform for open-pit mines is crucial for the management of mineral resources.

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With the rapid development of remote sensing technology and the continuous improvement in data acquisition ability [4], a substantial volume of high-resolution remote sensing imagery has been generated. These images offer robust support for various domains such as land resource management [5], [6], [7], urban development planning [8], [9], traffic management [10], and natural disaster monitoring [11], [12]. At the same time, the abundance of remote sensing data has also facilitated the monitoring of mineral resources. However, the unique imaging conditions of remote sensing imagery and the complex geographical environments of open-pit mining areas pose challenges for the task of remote-sensing-based open-pit mine detection. To improve detection accuracy, accurately analyzing the structural features of mining areas (such as edge, texture, shape, and spatial relationships) is crucial. By conducting an in-depth analysis of these structural features, key information about the morphology and distribution of objects within the mining area can be obtained, which enhances the interpretation accuracy of remote sensing data and improves the precision of mining area detection.

In the early years, the identification of open-pit mines could only be done by visual interpretation and manual delineation [13], which was costly, inefficient, and highly subjective [14]. Moreover, faced with the massive amount of remote sensing data from open-pit mining areas today, it is clearly inadequate. On the surface environment, different features of objects are reflected in remote sensing image data. The information from the red, green, blue, and near-infrared bands can provide abundant shallow features. Through mathematical statistical methods, the data features of the four bands within the mining area can be analyzed, so as to accurately identifying the scope of the open-pit mining area from the remote sensing image. With the development of machine learning, researchers have gradually applied various machine learning methods to open-pit mining remote sensing images [15], by continuously optimizing machine learning algorithms and integrating them with other methods to improve the accuracy of mine area detection. However, with the rapid growth of remote sensing data and the continuous improvement of task accuracy requirements, these traditional methods have become inadequate.

In recent years, with the development of deep learning, many classic convolutional neural networks (CNNs) have emerged in the field of image processing [10], [16], [17]. For instance, He et al. [18] proposed the residual network

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Fig. 1. Complex environment in open-pit mining area. (a) Interference from buildings within the mining area. (b) Influence of lighting conditions. (c) Blurred boundaries of the mining area.

(ResNet), which uses skip connections to alleviate the network degradation caused by the increase in network depth. Goodfellow et al. [19] introduced the generative adversarial network (GAN), which enhances the detection accuracy through the adversarial interplay between the generator and the discriminator. Sandler et al. [20] introduced MobileNet, which significantly reduces the number of parameters by replacing conventional convolutions with depthwise separable convolutions, thereby making deep learning more applicable in practical scenarios while maintaining accuracy. These models greatly promote the development of deep learning in the field of image processing by virtue of their powerful learning and modeling capabilities. With the introduction of the attention mechanism by Vaswani et al. [21], transformers, benefiting from their powerful long-range dependency modeling capabilities, have achieved unprecedented success in the field of natural language processing. Later, Dosovitskiy et al. [22] extended this mechanism to image processing by proposing the Vision Transformer (ViT). ViT takes full advantage of the transformer's ability to capture global features, demonstrating strong performance in tasks such as image classification, and offering greater flexibility and scalability compared with traditional CNNs. While these methods offer strong robustness and generalization, particularly in complex conditions, open-pit mining detection remains challenging due to environmental complexities and unclear boundaries (Fig. 1). Current models often rely on deepening network structures and increasing parameters to improve performance, but this results in overly complex models that are harder to train and deploy. Hence, there is an urgent need for a lightweight approach that balances fewer parameters, lower computational cost, and improved detection performance.

To address the above issues, we design a novel lightweight network called DASFNet. To achieve lightweight objectives, we select MobileNetV2 [20] as the backbone for preliminary feature extraction. Considering the relatively weaker feature extraction capability of MobileNet compared with backbones with a large number of parameters, we develop a neighbor feature aggregation module (NFAM) to integrate features from adjacent scales, enhancing the feature representation capability at each stage. In response to the challenge of defining the edges of open-pit mines in remote sensing, we design an edge feature compensation module (EFCM), which uses dilated convolutions with varying dilation rates to capture semantic information from different receptive fields in the feature map. Then, the extracted features are merged with the original features through residual connections to enhance the representation of mining area edge features. To facilitate the interaction and fusion of features across different scales, we adopt a stair fusion framework to merge adjacent feature maps. This framework consists of a cross-attention fusion module (CAFM), which effectively merges high-level features into low-level features by reweighting the input feature maps.

The main contributions of this article are as follows.

1) In this research, we design a novel lightweight detection network (only containing 3.14M parameters and costing 1.88 G FLOPs), called DASFNet, through adjacent feature aggregation and cross-attention fusion mechanism to address the challenges in remote sensing open-pit mine detection.
2) We develop an NFAM to integrate features from adjacent scales, which can enhance the feature representation ability at each stage.
3) To address the difficulty in defining the edges of open-pit mines, an EFCM with dilated convolutions with varying dilation rates is designed to capture semantic information from different receptive fields and enhance edge feature representation through residual connections.
4) A stepwise fusion framework consisting of CAFM is developed to merge adjacent feature maps, which effectively merge high-level features into low-level features by reweighting the input feature maps.
5) An new remote sensing open-pit dataset (CIGMR) is constructed. The CIGMR dataset comprises 3979 remote sensing images, each with a resolution of 256 × 256 pixels. To our knowledge, it is the first remote sensing open-pit mines dataset for detection.

The remainder of this article is organized as follows. Section II discusses the related work. Section III describes the proposed network architecture in detail. Datasets' introduction, experimental results, and analysis are presented in Section IV. The conclusions are given in Section V.

II. Related Work

A. Traditional Open-Pit Mining Area Detection Methods

In traditional open-pit mine detection methods, researchers primarily study the characteristics of remote sensing data and use the relationships between different spectral bands of mine area pixels to determine the mine area boundaries. For instance, Xie et al. [23] identified the mining area by calculating shallow features such as contrast, correlation, and homogeneity. With the development of machine learning, many researchers are applying machine learning methods to the task of open-pit mine detection. For example, Chen [24] used data from the GaoFen-1 (GF-1) satellite to study the application of several machine learning methods, including maximum likelihood estimation (MLE), minimum distance classification (MDC), and support vector machine (SVM), in the research of land use in mining areas. Maxwell et al. [25] combined random forest (RF) with feature-based filtering selection for land cover classification and mapping of open-pit mining areas and reclamation sites. Yu et al. [26] used deep learning to extract large-scale initial feature

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information, followed using RF to classify and extract the remaining feature information, to analyze land occupation in mining production. These methods are based on mathematical and machine learning methods, which use spectral information from different bands of remote sensing images to a certain extent, and improve the accuracy of classification of open-pit mining area. However, when faced with today's vast amounts of high-resolution remote sensing data with complex background environments, these traditional methods are difficult to achieve the expected results.

B. Deep-Learning-Based Open-Pit Mining Areas Detection Methods

Depth CNNs effectively extract features from local image areas through convolution, building a spatial hierarchy with multiple layers where lower layers capture basic features and higher layers form complex representations. Due to their strong performance in image processing, CNN-based methods are widely used. Since the introduction of fully convolutional networks (FCN) by Long et al. [27], several classic CNN-based semantic segmentation models have emerged, including SegNet [28], U-Net [29], ResU-Net [30], and U-Net++ [31], [32]. These methods have shown notable potential in medical image processing and autonomous driving. Similarly, these networks have been applied to open-pit mine detection. Xie et al. [2] proposed DUSegNet, which combines SegNet's pyramid structure with U-Net's skip connections to improve detection accuracy. Meng et al. [33] introduced GIPNet, incorporating the gather–inject–perceive (GIP) module to reduce information loss and the boundary perception (BP) module to enhance mine boundary detection accuracy. Ren et al. [34] used ResNet-34 to extract preliminary features and then applied an expectation maximization attention mechanism to learn important information from these features and optimized the initial prediction results using fully connected conditional random fields (FC-CRF) to enhance the extraction of open-pit mine boundaries. The HTDLTD method proposed by Feng et al. [35] purifies background information through low-rank tensor decomposition (LRTD), and then refines the coarse detection results using a spectral-distance-based approach to obtain the final detection results. Zhang et al. [36] proposed an object-oriented feature expression strategy that transforms pixel features into object features through segmentation, thereby reducing the impact of noise on detection results and enhancing detection performance.

The methods based on CNN have achieved relatively good performance to some extent, but the limited receptive field of convolutional operations makes it difficult to model long-range dependencies in image data, hindering the accurate capture of global contextual information and leading to a stagnation in detection accuracy. Although increasing the size of convolution kernel can enhance the receptive field, it will bring about a sharp increase in model parameters. In contrast, transformer, with its unique attention mechanism, has an obvious advantage in establishing long-range dependencies. Therefore, this article proposes a model that combines CNN with attention mechanisms to fully extract multiscale contextual information from image data. Subsequently, multiscale information is integrated to improve detection accuracy through the stepped fusion framework.

III. Methodology

In this section, we first present the overall framework of the proposed DASFNet. Then, we provide a detailed introduction to the NFAM, EFCM, and CAFM components. Finally, we discuss the loss function.

A. Overall Architecture

As shown in Fig. 2, DASFNet captures crucial information from remote sensing images, providing an efficient method for open-pit mine detection. We use MobileNetV2 as the backbone network for feature extraction, excluding global average pooling and the final fully connected layer. Each stage of the backbone includes a convolutional layer with a stride of 2, reducing the feature map size by half at each stage, eventually shrinking to one-thirty-second of the original resolution. This process produces five feature maps of different scales. For a balance between efficiency and parameters, we use only the feature maps from the last four stages. NAFM performs adjacent feature fusion on these maps, and spatial-channel attention enhances feature representation. EFCM uses dilated convolutions to extract features with various receptive fields, which are fused with the original features to improve edge information. Finally, CAFM fuses higher level features into lower level ones, restoring them to the first-stage resolution. The stair fusion architecture facilitates information interaction across all stages, enhancing detection accuracy.

B. Neighbor Feature Aggregation Module

Compared with some backbone networks with a higher number of parameters, MobileNetV2 has slightly less feature extraction capability. Typically, lower level features consist of low-level characteristics such as edges, textures, and points of objects, while higher level features contain high-level attributes such as object shapes and positions [37]. Given that features at different levels contain diverse information about the object, we opt to integrate multilevel features. This approach leverages detailed information from lower level features to complement feature specifics and uses high-level information from higher level features to enhance semantic understanding. Therefore, NFAM was designed to integrate features from different levels and enhance feature representation using spatial and channel attention, as depicted in Fig. 3.

NAFM is capable of receiving inputs from three different levels of features, as illustrated in Fig. 3. Since we only use the feature maps from stage 2 to 5, when performing adjacent feature aggregation operations for stages 2 and 5, we only use their adjacent higher level (stage 3) and lower level (stage 4) features. To facilitate feature fusion, we use a stride-2 max pooling operation and a 3×3 convolution on the low-level high-resolution features to align their size and dimensions with those of the intermediate features. As for the high-level low-resolution features, we use convolution operations and bilinear interpolation to ensure their alignment with the size and

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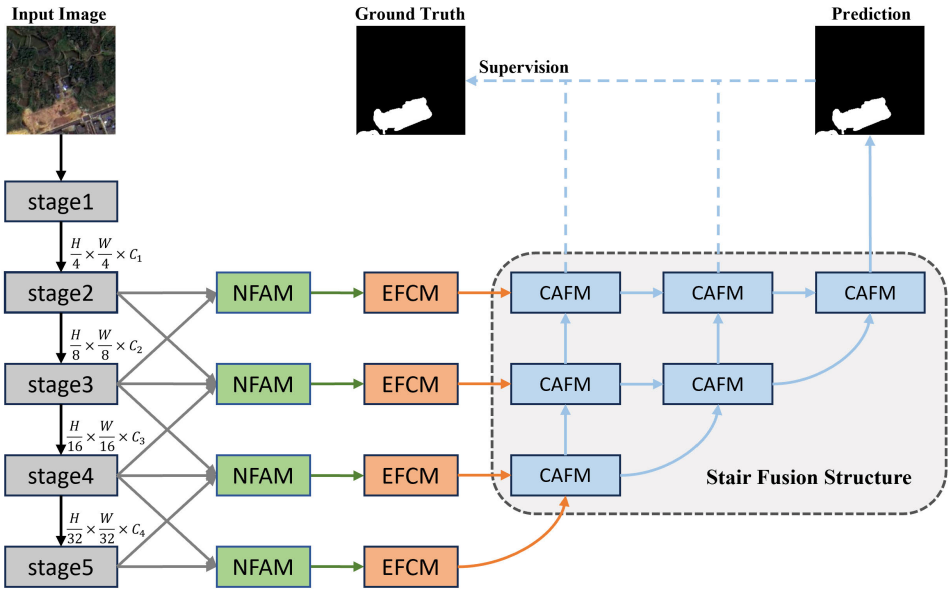


Fig. 2. Framework of DASFNet. The multiscale preliminary features of remote sensing image data are extracted by the lightweight MobileNetV2. NFAM is used to fuse the extracted adjacent multiscale features and enhance feature representation using spatial-channel attention. EFCM strengthens the edge features of mining areas by merging information from different receptive fields through dilated convolutions. CAFM is responsible for fusing information from adjacent scales to refine semantic detail.

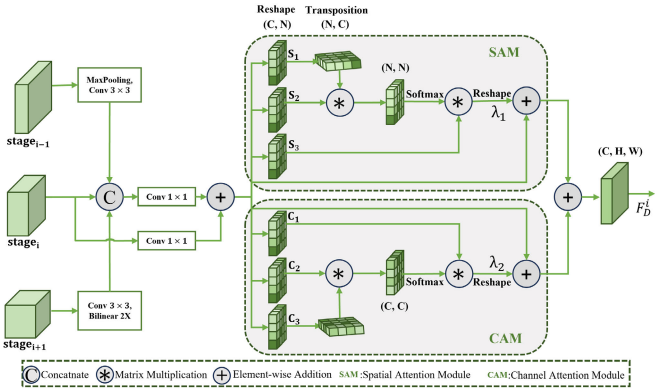


Fig. 3. Structure of NFAM.

dimensions of the intermediate features. Subsequently, we use concatenation to merge the three features across the channel dimension, followed by a 1×1 convolution for restoring the channel count. To ensure the prominence of the intermediate features, we incorporate a residual connection for combining them with the fused features. This process can be formulated as

$$\begin{aligned} F_1 &= \mathrm{Conv}_{3\times3}\big(\mathrm{MP}\big(\mathrm{stage}_{i-1}\big)\big) \\ F_2 &= \mathrm{Conv}_{1\times1}\big(\mathrm{stage}_i\big) \\ F_3 &= \mathrm{UP}\big(\mathrm{Conv}_{3\times3}\big(\mathrm{stage}_{i+1}\big)\big) \\ F_A &= F_2 + \mathrm{Conv}_{1\times1}\big(\mathrm{Cat}\big(F_1, \mathrm{stage}_i F_3\big)\big) \end{aligned} \tag{1}$$

where stage_{i-1}, stage_i, and stage_{i+1} represent the feature maps from three stages. $\mathrm{MP}(\cdot)$ denotes max pooling with a stride of 2. $\mathrm{Conv}_{1\times1}(\cdot)$ and $\mathrm{Conv}_{3\times3}(\cdot)$ represent the convolution operations with kernel sizes of 1 and 3, respectively. $\mathrm{UP}(\cdot)$ is the bilinear interpolation upsampling operation. $\mathrm{Cat}(\cdot)$ represents the concatenation along the channel dimension. F_A represents the preliminary fusion result of adjacent features by the NFAM module.

After the initial fusion of the three different levels of features, we apply spatial attention module (SAM) and channel attention module (CAM) to enhance the representation capability of the fused features.

1) Spatial Attention Module: SAM encodes contextual information into local features, enhancing the feature representation capability. First, we reshape the initially fused feature map $F_A \in \mathbb{R}^{C\times H\times W}$ three times to obtain three feature maps $S_1 = S_2 = S_3 \in \mathbb{R}^{C\times N}$, where C represents the number of channels, $N = H \times W$, and H and W are the height and width of the feature map, respectively. After performing matrix multiplication using S_2 and the transpose of S_1, a softmax layer is applied to obtain the spatial attention map $F_{\mathrm{SAM}} \in \mathbb{R}^{N\times N}$. Subsequently, matrix multiplication is performed on S_3 and F_{SAM}, followed by reshaping the result to $\mathbb{R}^{C\times H\times W}$. Finally, this result is multiplied by the spatial coefficient λ_1 and added elementwise to F_A to obtain the spatial-weighted feature map F_S. The specific calculation formula is as follows:

$$\begin{aligned} S_1, S_2, S_3 &= \mathrm{Reshape}(F_A) \\ F_{\mathrm{SAM}} &= \mathrm{Softmax}(S_2 * T(S_1)) \\ F_S &= F_A + \lambda_1 \cdot \mathrm{Reshape}(S_3 * F_{\mathrm{SAM}}) \end{aligned} \tag{2}$$

where $*$ represents matrix multiplication, $T(\cdot)$ denotes the transpose operation, and λ_1 is a learnable parameter in the network. F_S represents the processing result of the SAM module in the NFAM module.

2) Channel Attention Module: In high-level features, each channel's feature map can be considered as a manifestation of specific information. By implementing a CAM, we can enhance the interplay between channels to elevate the overall

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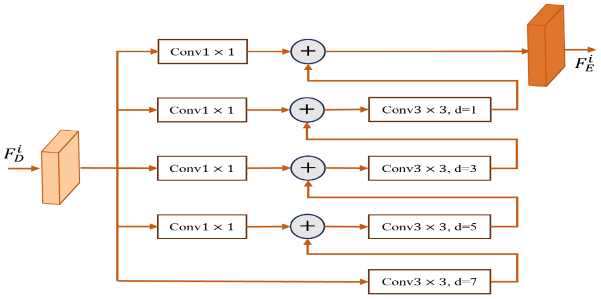


Fig. 4. Structure of EFCM.

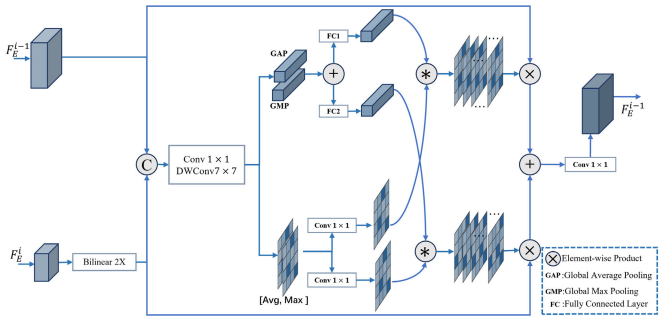


Fig. 5. Structure of CAFM.

semantic information representation capability of the features (Dual-attention). As depicted in Fig. 3, akin to the SAM operation, we reshape the input F_A into three maps $C_1 = C_2 = C_3 \in \mathbb{R}^{C\times N}$. Using matrix multiplication, we compute the product of C_2 and the transpose of C_1, followed by applying a softmax layer to obtain the channel attention map $F_{\text{CAM}} \in \mathbb{R}^{C\times C}$. Subsequently, we conduct matrix multiplication between F_{CAM} and C_3, and then reshape the result to $\mathbb{R}^{C\times H\times W}$. Afterward, we multiply it by the channel coefficient λ_2 and add it to F_A to derive the final channel-weighted feature map F_C. The calculation formula is as follows:

$$\begin{aligned} C_1, C_2, C_3 &= \text{Reshape}(F_A) \\ F_{\text{CAM}} &= \text{Softmax}(C_2 * T(C_1)) \\ F_C &= F_A + \lambda_2 \cdot \text{Reshape}(C_3 * F_{\text{CAM}}). \end{aligned} \tag{3}$$

F_C represents the processing result of the CAM module in the NFAM module.

Finally, the ultimate fused feature map is obtained through the combination of the outcomes from the SAM and CAM processes

$$F_D^i = F_S + F_C. \tag{4}$$

F_D^i represents the output result of the NFAM module after aggregating adjacent features in the ith stage.

C. Edge Feature Compensation Module

As the boundaries of open-pit mining areas typically transition gradually into nonmining regions, conventional methods face challenges in accurately defining their limits. Therefore, we have proposed EFCM, as illustrated in Fig. 4. We use dilated convolutions with four different dilation rates 7, 5, 3, 1 to extract information with varying receptive fields from the feature maps. Subsequently, residual connections are used to perform elementwise addition between the feature information captured from different receptive fields by dilated convolutions and the original feature map. This process aims to supplement the detail information in the original feature map with features from different receptive fields, enhancing the semantic representation capabilities of the features. Suppose that F_D^i represents the input of EFCM, the specific calculation formula is as follows:

$$\begin{aligned} F_{E1} &= \text{Conv}_{3\times3}^{d=7}(F_D^i) + \text{Conv}_{1\times1}(F_D^i) \\ F_{E2} &= \text{Conv}_{3\times3}^{d=5}(F_{E1}) + \text{Conv}_{1\times1}(F_D^i) \\ F_{E3} &= \text{Conv}_{3\times3}^{d=3}(F_{E2}) + \text{Conv}_{1\times1}(F_D^i) \\ F_E^i &= \text{Conv}_{3\times3}^{d=1}(F_{E3}) + \text{Conv}_{1\times1}(F_D^i) \end{aligned} \tag{5}$$

where $\text{Conv}_{3\times3}^{d=7}(\cdot)$ represents the 3×3 convolution operation with a dilation rate of 7, and F_E^i is the final output of EFCM.

D. Cross-Attention Fusion Module

In image detection tasks, decoding high-level features to the original image size is challenging because high-level features often lack fine details such as textures and corners, which are present in low-level features. To enhance detection accuracy, integrating features at various scales is crucial. To address this, we introduced CAFM, which reweights and sums feature maps using spatial and channel information to better combine high-level features with low-level details, as illustrated in Fig. 5.

The CAFM receives two feature inputs of different scales, denoted as F_E^{i-1} and F_E^i. First, the high-level feature F_E^i is upsampled using bilinear interpolation to match the resolution of F_E^{i-1}, and then the two features concatenated along the channel dimension. Subsequently, a 1×1 convolution is used to restore the number of channels, followed by a depthwise separable 7×7 convolution to aggregate the features from the concatenated input, as

$$f_c = \text{DWConv}_{7\times7}\left(\text{Conv}_{1\times1}\left(\text{Cat}\left(F_E^{i-1}, \text{UP}\left(F_E^i\right)\right)\right)\right) \tag{6}$$

where $\text{DWConv}_{7\times7}(\cdot)$ represents the depthwise separable 7×7 convolution. Subsequently, two separate branches are used to obtain the channel information sequences and spatial information maps of f_c. In the first branch, global average pooling and global max pooling are applied to f_c, generating two sequences containing the channel information. These sequences are then added elementwise together and followed by two different fully connected operations, resulting in the reconstructed sequences seq1 and seq2. In the second branch, max pooling and average pooling are performed along the channel dimension of f_c, resulting in two maps containing different spatial information. These maps are combined through a concatenation operation, and then undergo two different 1×1 convolutions to produce two maps containing spatial information: $m1$ and $m2$

$$\begin{aligned} \text{seq1}, \text{seq2} &= \text{MLP}_i(\text{GAP}(f_c) + \text{GMP}(f_c)) \\ m1, m2 &= \text{Conv}_{1\times1}^i(\text{Cat}(\text{MAX}(f_c), \text{MEAN}(f_c))) \\ & \qquad i = 1, 2 \end{aligned} \tag{7}$$

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where GAP(·) and GMP(·) represent global average pooling and global max pooling, respectively. MAX(·) and MEAM(·) represent max pooling and average pooling operations along the channel dimension, respectively. MLP is a multilayer perceptron.

We conduct matrix multiplication between seq1 and $m1$, as well as between seq2 and $m2$, resulting in two weight matrices that integrate spatial and channel information. These weight matrices are then used to weigh F_E^{i-1} and F_E^i, respectively, followed by aggregating the weighted features to obtain the integrated feature map. Subsequently, a 1×1 convolution is used on the outcome to generate the ultimate integrated feature. The calculation formula is as follows:

$$F_E^{i-1} = \mathrm{Conv}_{1\times1}\big((\mathrm{seq1} * m1) \cdot F_E^{i-1} + (\mathrm{seq2} * m2) \cdot \mathrm{UP}\big(F_E^i\big)\big) \tag{8}$$

where F_E^{i-1} represents the result of the progressive fusion of F_E^i and F_E^{i-1} through the CAFM module.

We arrange the CAFM in a cascading structure, enabling comprehensive fusion of features at each level. This ensures that the final feature map contains relatively complete and accurate semantic information, thereby enhancing detection accuracy.

E. Loss Function

In the task of identifying open-pit mining areas in remote sensing images, there is often a class imbalance problem due to the nonmining areas covering a much larger area than the mining areas in the samples. The use of only the cross-entropy loss function for calculation results in minor fluctuations in the loss value, which can impact the convergence speed of the model. To address this issue, we have adopted a hybrid approach that combines binary cross-entropy (BCE) loss and Dice loss. The calculation formulas for BCE and Dice loss are as follows:

$$\mathrm{Loss}_{\mathrm{BCE}} = \frac{1}{H \times W} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw})$$
$$l(P_{hw}, Y_{hw}) = P_{hw} \cdot \log Y_{hw} + (1 - P_{hw}) \cdot \log(1 - Y_{hw}) \tag{9}$$
$$\mathrm{Loss}_{\mathrm{Dice}} = \frac{1}{H \times W} \sum_{h=1,w=1}^{H,W} \left(1 - \frac{2 \cdot P_{hw} \cdot Y_{hw}}{\|P_{hw}\| + \|Y_{hw}\|}\right) \tag{10}$$

where P and Y represent the predicted values and true labels, respectively, and (h, w) is the pixels location, respectively. $l(\cdot)$ represents the cross-entropy loss. $\|\cdot\|$ denotes the l_1-norm.

The results of the three lowest level CAFMs in the model encompass all the information from the image data. Consequently, we calculate the loss function for each of these three results, as depicted in Fig. 2. Subsequently, by assigning different weights to these three losses and summing them together, we derive the final loss of the model using the following calculation formula:

$$\mathrm{Loss} = \mathrm{Loss}_{\mathrm{BCE}} + \mathrm{Loss}_{\mathrm{Dice}}$$
$$\mathrm{Loss}_{\mathrm{final}} = w_1\mathrm{Loss}_1 + w_2\mathrm{Loss}_2 + w_3\mathrm{Loss}_3 \tag{11}$$

TABLE I
BASIC INFORMATION OF CIGMR DATA

Data Source	Bands	Resolution	Data Depth	Data Format
Gaofen-1	R/G/B/NIR 4 Bands	2.0m	8-bit	TIFF
Gaofen-2	R/G/B/NIR 4 Bands	0.8m	8-bit	TIFF
Gaofen-6	R/G/B/NIR 4 Bands	2.0m	8-bit	TIFF
Gaofen-7	R/G/B/NIR 4 Bands	0.7m	8-bit	TIFF
ZY-1	R/G/B/NIR 4 Bands	2.0m	8-bit	TIFF
ZY-3	R/G/B/NIR 4 Bands	2.0m	8-bit	TIFF
CBERS-4	R/G/B/NIR 4 Bands	2.0m	8-bit	TIFF

where Loss_1, Loss_2, and Loss_3 represent the losses calculated from the three lowest level CAFMs, and w_1, w_2, and w_3 are the corresponding weights assigned to each loss.

IV. EXPERIMENTS

Our model, implemented using the PyTorch framework, is trained on a single NVIDIA Tesla A40 GPU. During the training phase, we initialize the backbone of DASFNet with parameters from a MobileNetV2 [21] model pretrained on ImageNet; the remaining network parameters are initialized according to the method detailed in [38]. For model optimization, we use the stochastic gradient descent (SGD) algorithm with a momentum of 0.99 and a weight decay parameter of 0.0005. The learning rate is initially set to 0.0001 and decreased to 0.000001 using a cosine decay function over 200 epochs. The batch size is set at 32. Following each training cycle, we validate the model on a separate validation set and use the best-performing model for final evaluation on the test set.

A. Dataset Settings

Due to the limited availability of high-quality public datasets in the field of remote sensing open-pit mine detection, there is a scarcity of current deep-learning-based methods for this task, and their performance is subpar. To develop an effective model for open-pit mine detection, we collaborate with relevant organizations to create a detection dataset named CIGMR, which consists of 3979 images and corresponding binary label images. The dataset has been divided into training, validation, and testing sets in an 8:1:1 ratio, resulting in 3183 images for training, 398 images for validation, and 398 images for testing.

Moreover, since open-pit mining areas are often in wilderness or forested regions, they are frequently shrouded in fog during autumn and winter, which complicates mining site detection. To test our model's performance in such conditions, we processed the CIGMR dataset using a center point synthesis method based on a standard optical model, creating the CIGMR(fog) dataset to simulate foggy conditions [10]. This dataset mirrors the original CIGMR dataset in image sources and data division, with the only addition being the fog effect.

1) Data Description: CIGMR has selected remote sensing images captured within Chongqing China from 2021 to 2023 as the data source. The most prevalent types of open-pit mines in Chongqing, including limestone (such as decorative limestone, construction stone limestone, cement limestone), sandstone (such as flagstone, construction sandstone, cement

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Fig. 6. Sample images from the CIGMR and CIGMR(fog) datasets. First row: original images. Second row: fogged images. Third row: corresponding labels.

additive sandstone), shale (such as brick and tile shale, ceramsite shale), and other types (such as fluorite, calcite), are chosen as the mining identification objects. The data for CIGMR are described in Table I.

2) Dataset Construction: We used ArcGIS software to manually interpret remote sensing satellite data, delineating open-pit mining boundaries and generating vector annotation files, a process conducted by professional technicians. Based on the location data, we extracted regions containing open-pit mining areas from the images and standardized them to 256 × 256 pixels for model training. Due to uneven color distribution in the original remote sensing images, we applied percentage truncation stretching to normalize pixel values across all bands from 0 to 255. In total, we collected 3979 image patches, each containing at least one mining area. Examples are shown in Fig. 6.

B. Data Augmentation Technology

To enrich the dataset, prevent overfitting during the training process, and improve the model's generalization capability, we used a series of data augmentation techniques. These techniques not only enhance the diversity of the data but also assist the model in adapting to various potential real-world scenarios, thereby improving its performance on unseen data.

1) Random Flipping: This method flips images either horizontally or vertically, allowing the model to learn orientation-invariant features and adapt to various image perspectives.

2) Random Rotation: Images are rotated by 0°, 90°, 180°, or 270°, providing diverse orientations of the same image and helping the model learn rotation-invariant features.

3) Random Scaling and Cropping: Images are randomly rescaled (by 1–1.2 times) and cropped to 256 × 256 pixels. This process introduces variations in object sizes and ensures the model learns to focus on different regions of the image.

4) Random Gaussian Blurring: A Gaussian blur filter is applied with a randomly selected radius (0–1), introducing variations in sharpness and simulating different focus or lens effects.

C. Evaluation Metrics

To evaluate the performance of the proposed model, we use the primary evaluation metrics commonly used in the detection field. These include precision (Pre.), recall (Rec.), intersection over union (IoU), $F1$-score ($F1$), and overall accuracy (OA) [10]. The values for these five metrics range from 0 to 1, with higher values indicating better network performance.

D. Comparative Methods

To showcase the advanced nature of our model, we compared DASFNet with six state-of-the-art (SOTA) methods in remote sensing image detection and semantic segmentation: U-Net [29], SegNet [28], DUSegNet [2], GIPNet [33], ASNet [39], and EasyNet [40]. These methods are prominent in the field. Extensive experiments on the CIGMR and CIGMR(fog) datasets, detailed in Table II, show that our method balances performance with parameter count effectively.

E. Comparative Experiments

Table II presents the performance comparison results of our method and other benchmark methods on the CIGMR and CIGMR(fog) test sets. By comparing the values of five evaluation metrics, model parameters, and FLOPs, the superiority of our model is comprehensively demonstrated. Figs. 7 and 8 present the visual comparison results of our model with other SOTA methods on two datasets, respectively. In these figures, TP, TN, FP, and FN are highlighted by white, black, red, and green, respectively.

1) Detection in CIGMR Dataset: As shown in Table II (CIGMR dataset), our model achieves an excellent balance between performance and lightweight design, with the fewest parameters while maintaining top scores across all five evaluation metrics. Notably, our $F1$-score is 1.03% higher than EasyNet, and IoU is 1.62% higher than GIPNet.

In Fig. 7(a)–(e), each subfigure represents detection results from different models. In Fig. 7(a), our model accurately detects target areas despite interference from nearby buildings, while other methods show false positives or misses. In Fig. 7(b), our model effectively distinguishes targets with colors similar to the background, unlike U-Net, SegNet, and GIPNet, which mistakenly detect nonmining areas. Fig. 7(c) shows ASNet and EasyNet misclassify roads as mining areas, but our model avoids this. Fig. 7(d) highlights our model's performance on large targets. Finally, in Fig. 7(e), our EFCM accurately captures complex boundaries, outperforming other methods.

2) Detection in CIGMR(Fog) Dataset: Table II right (CIGMR(fog) dataset) shows that our model demonstrates superior performance in foggy conditions, achieving the best results across all metrics compared with other SOTA methods. Specifically, when handling detection tasks in foggy environments, our model experiences only a slight performance drop compared with normal conditions, with precision (Pre) and $F1$-score decreasing by just 0.52% and 0.27%, respectively. In contrast, the other six methods show a drop in Pre and $F1$ exceeding 1%, particularly for U-Net and SegNet, which are more traditional detection methods. Their $F1$-scores

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TABLE II

COMPARISON RESULTS OF FIVE EVALUATION INDICATORS (PRE., REC., $F1$, IOU, AND OA) ON THE CIGMR AND CIGMR(FOG) DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

Models	Params. (M)	FLOPs (G)	CIGMR Dataset					CIGMR(fog) Dataset				
			Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
U-Net	28.75	13.99	86.55	74.38	80.00	74.68	97.21	83.69	73.26	77.11	71.70	96.22
SegNet	1.63	0.64	88.99	87.21	85.79	75.12	97.82	85.89	85.20	83.72	73.32	96.58
DUSegNet	30.55	14.60	89.24	89.37	88.31	80.68	97.94	85.44	86.31	86.85	77.48	96.82
GIPNet	26.90	12.27	91.53	88.86	89.17	82.10	98.01	90.13	87.23	86.02	81.31	97.11
ASNet	65.40	20.10	91.62	89.85	90.04	81.89	98.00	90.12	86.84	88.74	80.16	97.34
EasyNet	3.90	4.32	90.51	89.72	90.11	82.00	98.12	88.33	87.62	88.33	81.08	97.52
DASFNet	3.14	1.88	92.41	89.90	91.14	83.72	98.53	91.89	89.20	90.87	83.51	98.41

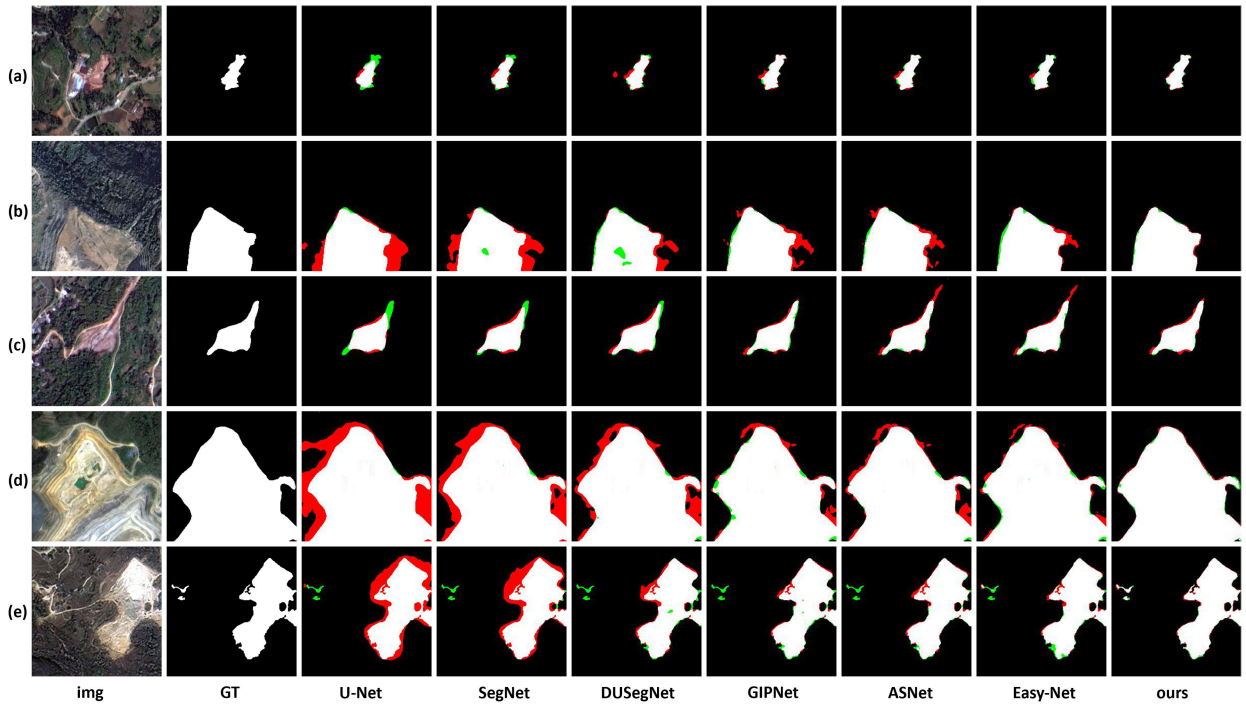


Fig. 7. Visualization performance of different methods on the CIGMR dataset. (a)–(e) Detection results of different methods on the CIGMR dataset. TP, TN, FP, and FN are highlighted by white, black, red, and green, respectively.

decrease by 2.89% and 2.07%, respectively. This demonstrates that our model has a stronger resistance to interference and maintains high performance even in challenging environments.

As shown in Fig. 8, foggy images blur the edges of mining areas, making them harder to distinguish. The other six SOTA methods show numerous false detections, particularly along the edges. In contrast, our method, enhanced by EFCM and random Gaussian blur data augmentation, more accurately defines the blurred boundaries. The false detection areas (in red and green) are much smaller with our model, demonstrating its strong robustness in handling both normal and foggy conditions.

3) Comparison of Model Efficiency and Performance: Fig. 9 presents a comparison of our method with six other SOTA methods in terms of model performance ($F1$-score), model parameters (params), and computational cost (FLOPs). From the figure, it is evident that while some methods achieve high detection performance, they come with significant drawbacks in terms of large parameter sizes and substantial computational overhead. For instance, ASNet achieves impressive results with $F1$-scores of 90.04% and 88.74% on the CIGMR and CIGMR(fog) datasets, respectively, but it requires 65.4M parameters and 20.1 G FLOPs. This situation inevitably makes these methods challenging to deploy on devices with limited energy resources and computational capabilities, such as drones and satellites, hindering their real-world applications. In contrast, our model maintains low parameter sizes and computational costs while ensuring competitive detection performance.

F. Ablation Studies

We conduct comprehensive ablation experiments to validate the effectiveness of each module in our proposed model and to understand the relationship between different backbone models in terms of parameters and performance. By gradually removing different modules from the model and observing the changes in performance, we are able to assess the contribution of each module to the overall performance of the model.

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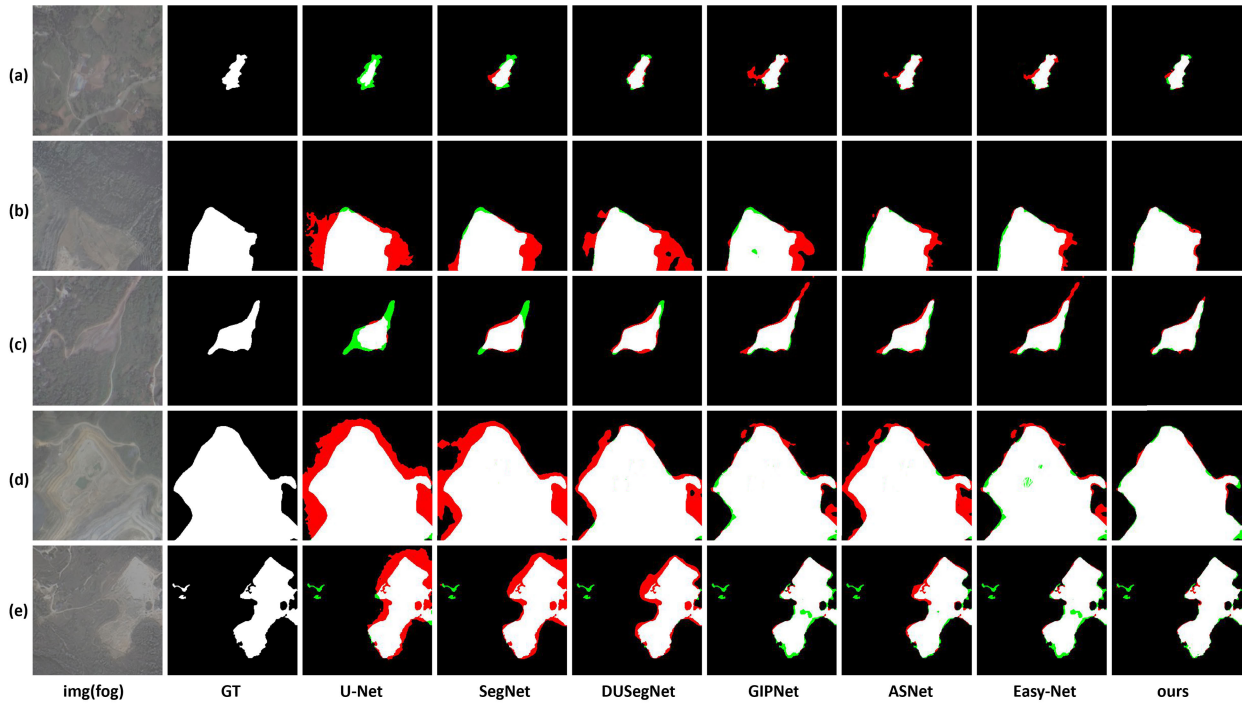


Fig. 8. Visualization performance of different methods on the CIGMR(fog) dataset. (a)–(e) Detection results of different methods on the CIGMR(fog) dataset. TP, TN, FP, and FN are highlighted by white, black, red, and green, respectively.

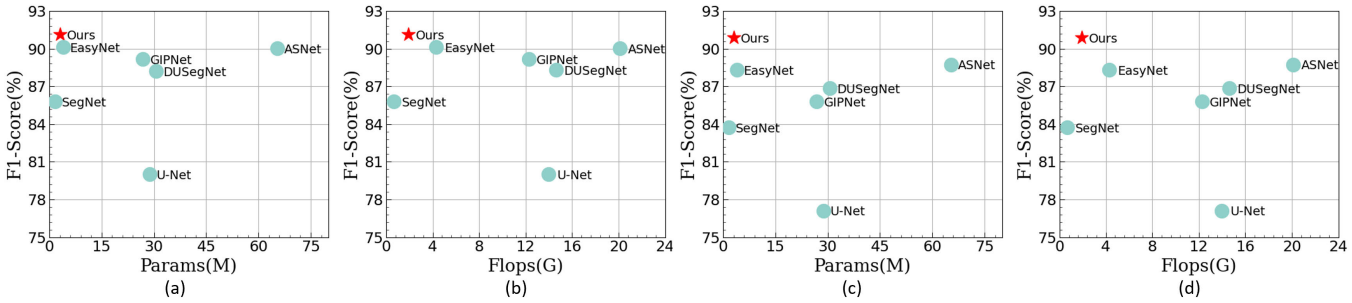


Fig. 9. Model complexity comparison of different methods in terms of $F1$-score, Params (memory cost), and FLOPs (computational cost), on the CIGMR and CIGMR(fog) datasets, respectively. (a) and (b) CIGMR dataset. (c) and (d) CIGMR(fog) dataset.

1) Effectiveness of Different Backbones: To evaluate the impact of different backbones on our model's performance and parameter count, we tested various feature extraction networks [Table III(a)]. Thanks to NFAM's multiscale feature aggregation, the MobileNet backbone achieves the lowest parameter count and FLOPs while maintaining similar performance. Compared with ResNet18 and VGG16, the metrics are closely aligned, and when compared with the top-performing ResNet34, the $F1$-score drops by only 0.2% and 0.11% across the two datasets.

2) Effectiveness of NFAM: In our DASFNet, NFAM aggregates multiscale features from the backbone. To assess NFAM's effectiveness, we conducted an experiment where it was removed, allowing EFCM to directly process MobileNet's features (Table III, #5). On the CIGMR dataset, removing NFAM led to a 1.42% drop in $F1$-score, 0.87% in IoU, and 2.02% in recall, proving NFAM's value. On the CIGMR(fog) dataset, $F1$ dropped by 2.23%, showing NFAM's ability to mitigate fog-related issues. In addition, when only the first half of NFAM was used (Table III #6), performance slightly decreased across all metrics, confirming the importance of the spatial-channel attention mechanism.

3) Effectiveness of EFCM: We use the EFCM module to integrate features from various receptive fields, enhancing target edge representation. To assess its impact, we remove EFCM, resulting in a w/o-EFCM version (#7 in Table III), which shows Pre. drops by 4.08% and 5.51% on the two datasets. Replacing the dilated convolution in EFCM with a regular 3×3 convolution (#8) confirms that progressive fusion from large to small receptive fields enhances semantics. In addition, removing residual connections (#9) leads to a significant performance drop, as multiple convolutions without residuals result in substantial information loss across all metrics.

4) Effectiveness of CAFM: We use the CAFM module to reweight and aggregate high-level features into low-level features. To verify its effectiveness, we replaced CAFM with a standard 3×3 convolution, where high-level features are

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TABLE III

QUANTITATIVE COMPARISON ON THE CIGMR AND CIGMR(FOG) DATASETS TO EVALUATE THE IMPACT OF BACKBONE, LOSS FUNCTIONS, FUSION STRUCTURE, AND EACH MODULE ON THE PERFORMANCE OF OUR PROPOSED MODEL

No.	Versions	Efficiency		CIGMR Dataset					CIGMR(fog) Dataset				
		Params.(M)	FLOPs(G)	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
#1	DASFNet	3.14	1.88	92.41	89.90	91.14	83.72	98.53	91.89	89.20	90.87	83.51	98.41
(a) Different Backbones													
#2	ResNet18	17.50	8.90	91.36	89.32	90.86	83.45	98.25	90.41	88.74	89.84	82.12	98.01
#3	ResNet34	26.43	10.21	92.67	90.71	91.34	84.07	98.77	91.26	89.20	90.98	83.52	98.15
#4	VGG-16	172.21	17.61	91.86	90.12	90.36	83.43	98.31	90.43	89.10	89.74	82.72	97.93
(b) Neighbor Feature Aggregation Module (NFAM)													
#5	w/o NFAM	2.72	1.67	89.13	87.88	89.72	82.85	97.51	88.41	85.93	87.49	82.42	97.23
#6	w/o Dual-Att	3.12	1.80	91.53	89.15	90.84	83.43	98.37	89.31	88.44	89.17	82.84	97.55
(c) Edge Feature Compensation Module (EFCM)													
#7	w/o EFCM	2.51	1.04	88.33	86.96	89.04	82.42	97.34	86.38	84.82	87.18	81.82	97.06
#8	dilation:0	3.14	1.88	91.17	88.31	89.96	83.12	98.11	91.51	86.84	88.19	83.03	97.73
#9	w/o residual	3.10	1.80	90.07	86.83	87.26	81.92	97.15	89.41	86.27	86.94	81.42	97.01
(d) Cross Attention Fusion Module (CAFM)													
#10	w/o CAFM	2.86	1.35	89.24	87.37	88.94	82.68	97.46	88.31	86.80	87.14	81.40	97.03
(e) Stair Fusion Structure													
#11	w/o SFS	2.86	1.35	92.24	89.37	90.94	83.58	98.46	91.55	89.73	90.31	83.32	98.23
(f) Loss Function													
#12	BCE	3.14	1.88	92.33	89.58	90.81	83.55	98.47	91.27	89.51	90.54	83.41	98.26
#13	Dice	3.14	1.88	92.14	89.12	90.95	83.37	98.33	91.41	88.87	90.40	83.52	98.13

directly added after upsampling. As shown in Table III #10, removing CAFM led to a significant performance drop on the CIGMR and CIGMR(fog) datasets, with precision decreasing by 3.17% and 3.58%. This confirms the effectiveness of CAFM.

5) Effectiveness of Stair Fusion Structure: The stair fusion structure enables more comprehensive integration of features at each level. To verify the effectiveness of this framework, we design a model denoted as w/o SFS, as shown in Table III #11. This model removes the final three CAFM modules, retaining only the initial fused features for loss calculation with the labels. The results indicate that the stair fusion structure can better integrate multiscale features, enhancing the expression of semantic information and improving model effectiveness.

6) Effectiveness of Hybrid Loss: To achieve better model convergence, we use a hybrid loss function incorporating Dice loss and BCE loss. To verify its effectiveness, we compare our hybrid approach with methods using only BCE loss or only Dice loss. As shown in Table III(f), the hybrid loss method outperforms the methods that use either Dice loss or BCE loss alone.

V. CONCLUSION

In this work, we design a novel lightweight network for detecting open-pit mining areas. First, we introduce the NFAM to aggregate adjacent features from the multiscale features extracted by the backbone. Next, we design the EFCM module to merge features from different receptive fields, enhancing the expression of target edge information. Finally, we develop the CAFM module and stacked it in a stair fusion structure to better integrate features at various levels and strengthen the semantic information of the final feature map. Extensive experiments demonstrate that our model is superior to other SOTA methods on our constructed CIGMR and CIGMR(fog) datasets. To our knowledge, this dataset is the first remote sensing open-pit detection dataset.

This model has already been applied in practical production activities and has demonstrated impressive performance, significantly improving work efficiency. However, there is still a certain rate of false detections, where nonmining areas such as bare soil or uncultivated farmland are occasionally misidentified as mining areas. To address these issues, we plan to continue expanding and enriching our dataset in future work, including adding nonmining images that resemble mining areas as negative samples to enhance the model's ability to solve the aforementioned problems. In addition, we will consider how to further enhance the model's real-time performance and evaluate whether the model demonstrates excellent transfer learning capability when applied to unfamiliar datasets.

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